Next Generation Data Centers and Energy Management Systems

Lecture 1B: The Modern Data Center Ecosystem; Thermodynamics

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Fall 2026

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Apply first and second law analysis to a data center rack
- Analyze a data center cooling tower using psychrometric principles

Typical layout of a data center facility

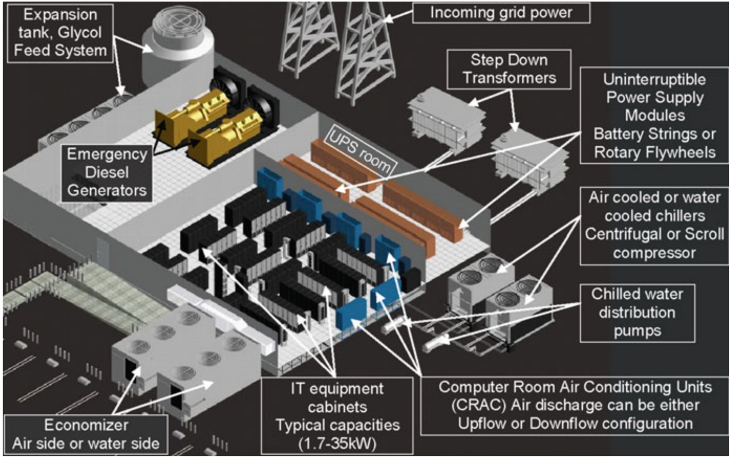

Figure 1: Y. Joshi and P. Kumar. Introduction to Data Center Energy Flow and Thermal Management, 2012.

Hot-aisle/cold-aisle layout

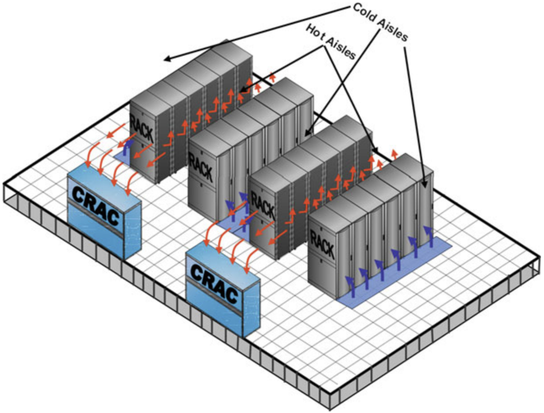

Figure 2: Y. Joshi and P. Kumar. Introduction to Data Center Energy Flow and Thermal Management, 2012.

Thermodynamics of information

- Almost all electrical energy consumed by a server is converted to heat: Joule heating.
- The goal is to do bit-shifting, which is low entropy; however, the byproduct is massive entropy generation in the form of low-grade heat.

Cooling infrastructure: Psychrometric principles

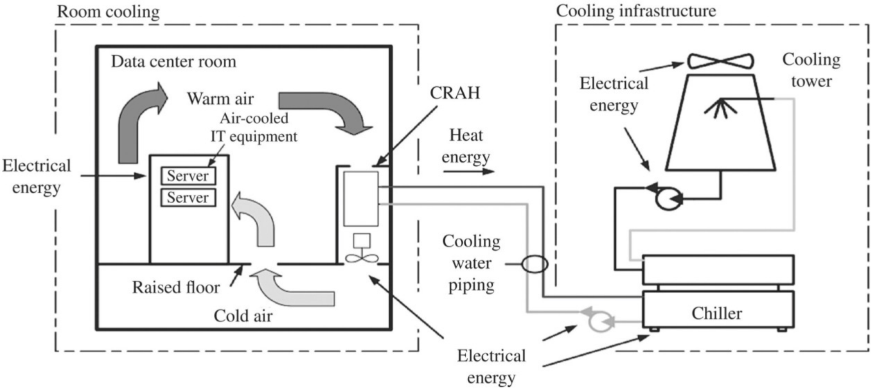

Figure 3: H. Geng. Data Center Handbook. 2nd Edition, Wiley, 2021. Chapter 14. Example: psychrometricPrinciples.ipynb